

D872

Secondary Chemistry Curriculum

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher candidate's phase-change unit starts the week after spring break, and the department just bought new hot plates. She wants to know what her learners think the bubbles in boiling water are made of. What should she do first?

- A.** Lecture on vaporization and correct misconceptions if any come up.
- B.** Run a boiling demo on the new hot plates, since hands-on work always fixes misconceptions.
- C.** Give a short probe asking learners to explain what the bubbles contain, then sort their answers (air, hydrogen and oxygen, water vapor).
- D.** Give a graded vocabulary quiz on phase-change terms.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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2.

Exit tickets printed on half-sheets of recycled paper show that 60% of a class set up a limiting-reactant problem incorrectly. The unit test is in nine days. What is the best use of this result?

- A.** Reteach the setup the next day with a targeted mini-lesson and re-check.
- B.** Stop using exit tickets; they never show more than attendance.
- C.** Record the scores as a unit grade.
- D.** Move on, because 40% understood it.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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3. [TRANSFER — reversed stem / different variable]

Which of these is a summative assessment?

- A.** A warm-up question answered on whiteboards
- B.** A one-on-one conference during a lab
- C.** A traffic-light self-rating of confidence
- D.** An end-of-unit exam scored for the report card

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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4.

A teacher candidate compares two free online simulations of a heated solid. Simulation A is plain and shows particles vibrating faster and spreading slightly apart. Simulation B has bright graphics, a game score, and shows the particles themselves growing larger as the temperature rises. Which should she choose?

- A. Neither, because simulations are always less accurate than labs.
- B. Simulation B, because learners will be more engaged by it.
- C. Both, shown back to back without comment.
- D. Simulation A, because it shows motion and spacing changing, not particle size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A textbook animation shows electrons circling the nucleus in fixed, planet-like orbits. How should a teacher use it?

- A. Never use any model that has limitations.
- B. Replace it with a vocabulary list about orbitals.
- C. Use it for energy levels, then explicitly discuss that electrons do not follow fixed paths.
- D. Present it as an accurate picture of the atom.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A lesson opens with a demonstration, from a kit that cost \$60, in which a clear solution suddenly turns deep blue. Learners are curious and full of questions. Following the 5E model, what should happen next?

- A. A quiz on the demonstration.
- B. Learners investigate variables that change the timing of the color change.
- C. Learners apply the idea to a new industrial process.
- D. The teacher explains the reaction mechanism and rate laws.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A class of 32 learners, taught in a room with a flickering projector, is new to multistep grams-to-grams stoichiometry, and most are stuck on where to start. Which sequence is best supported by research on novice learners?

- A. Show only the final answers so learners can check their work.
- B. Worked examples, then partly completed problems, then independent practice.
- C. Thirty independent problems first, with examples afterward.
- D. Never show examples, so learners discover the method themselves.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A teacher candidate finds a density lab online with a step-by-step procedure, a completed data table, and the expected answer. It was posted by a popular teaching blog with thousands of followers. She wants to turn it into guided inquiry. What change does that?

- A. Give the question “Which unknown metal is this?” and have learners design their own procedure.
- B. Share the expected answer first so learners can confirm it.
- C. Let learners investigate any question they choose, with no guidance.
- D. Keep the step-by-step procedure but remove the expected answer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In a lab, the teacher provides the question and the procedure, and learners do not know the expected result. What level of inquiry is this?

- A. Confirmation inquiry
- B. Guided inquiry
- C. Open inquiry
- D. Structured inquiry

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

To check understanding, a teacher candidate asks learners to draw what salt (NaCl) looks like dissolved in water. The class had just finished a lab using iodized table salt from a grocery store. Which drawing shows correct understanding?

- A. Separate Na^+ and Cl^- ions spread among water molecules, each surrounded by water
- B. Neutral Na and Cl atoms spread among water molecules
- C. Intact NaCl pairs floating among water molecules
- D. Na^+ and Cl^- ions piled at the bottom of the container

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which drawing correctly shows table sugar (sucrose) dissolved in water?

- A. Only water molecules, since the sugar has disappeared
- B. Separate carbon, hydrogen, and oxygen atoms
- C. Whole sucrose molecules spread evenly among water molecules
- D. Sucrose split into positive and negative ions

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

The lesson objective reads: “Learners will predict the products of double-replacement reactions using solubility rules.” The candidate’s mentor prefers short assessments that fit on one page. Which assessment task aligns best?

- A. Have learners define “double-replacement reaction.”
- B. Give reactant pairs and have learners predict the products, name any precipitate, and cite the rule used.
- C. Only a full lab practical can ever measure prediction.
- D. Have learners name ten ionic compounds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher candidate's phase-change unit ~~starts the week after spring break, and the department just bought new hot plates.~~ She wants to know what her learners think the bubbles in boiling water are made of. What should she do first?

Target word: first instructional move **Decides:** Prior ideas must be surfaced before teaching.

A	Dropped element	Skips eliciting, so hidden misconceptions stay hidden.
B	Absolute language	Activities without sensemaking can leave misconceptions intact.
C	Correct	Eliciting ideas lets her plan instruction that confronts specific misconceptions.
D	Adjacent category	A summative recall check does not reveal learners' explanations.

2. Answer A Concept 2

Exit tickets ~~printed on half sheets of recycled paper~~ show that 60% of a class set up a limiting-reactant problem incorrectly. ~~The unit test is in nine days.~~ What is the best use of this result?

Target word: formative use of data **Decides:** The results arrive mid-learning and show a specific gap.

A	Correct	Formative data should change instruction while learning is still happening.
B	Absolute language	This one just revealed a clear learning gap.
C	Adjacent category	Treats a formative check as summative.
D	Inverted operation	Reads the minority result as the signal.

3. Answer D Concept 2 · transfer

Which of these is a summative assessment?

Target word: identify summative by purpose **Decides:** Summative judges learning at the end.

A	Inverted operation	A quick check used to adjust the lesson is formative.
B	Confusion set	Real-time feedback during learning is formative.
C	Adjacent category	Self-assessment during learning guides next steps; it is formative.
D	Correct	It judges achievement after instruction.

4. Answer D Concept 3

A teacher candidate compares two free online simulations of a heated solid. Simulation A is plain and shows particles vibrating faster and spreading slightly apart. Simulation B ~~has bright graphics, a game score, and~~ shows the particles themselves growing larger as the temperature rises. Which should she choose?

Target word: choose the scientifically accurate resource **Decides:** Particles do not grow when heated; spacing and motion change.

A	Absolute language	A well-built simulation can show particle behavior no lab can.
B	Camouflage	Engagement does not outweigh a science error.
C	Dropped element	Leaves the inaccurate model unchallenged.
D	Correct	B reinforces the misconception that particles themselves expand.

5. Answer C Concept 3 • transfer

A textbook animation shows electrons circling the nucleus in fixed, planet-like orbits. How should a teacher use it?

Target word: using a limited model responsibly **Decides:** Models are useful but must be framed with their limits.

A	Absolute language	Every model has limits; that would remove all models.
B	Adjacent category	Drops the visual model instead of teaching with it critically.
C	Correct	Naming a model's limits builds accurate understanding.
D	Confusion set	Teaches the planetary-orbit misconception.

6. Answer B Concept 4

A lesson opens with a demonstration, ~~from a kit that cost \$60,~~ in which a clear solution suddenly turns deep blue. Learners are curious and full of questions. Following the 5E model, what should happen next?

Target word: next 5E phase after Engage **Decides:** Explore comes before Explain.

A	Dropped element	Jumps to Evaluate and skips the learning phases.
B	Correct	The Explore phase lets learners build experience before formal explanation.
C	Adjacent category	That is Elaborate, which comes after Explain.
D	Inverted operation	Explain before Explore reverses the sequence.

7. Answer B Concept 5

A class ~~of 32 learners, taught in a room with a flickering projector,~~ is new to multistep grams-to-grams stoichiometry, and most are stuck on where to start. Which sequence is best supported by research on novice learners?

Target word: best practice sequence for novices **Decides:** Novices need worked examples before independent practice.

A	Dropped element	Leaves out the solution steps learners need to see.
B	Correct	Fading lowers cognitive load and shifts work to learners over time.
C	Inverted operation	Reverses the sequence and overloads novices.
D	Absolute language	Unguided discovery is inefficient for novices on complex procedures.

8. Answer A Concept 6

A teacher candidate finds a density lab online with a step-by-step procedure, a completed data table, and the expected answer. It was ~~posted by a popular teaching blog with thousands of followers.~~ She wants to turn it into guided inquiry. What change does that?

Target word: move a lab to guided inquiry **Decides:** Guided: teacher gives the question; learners design the procedure.

A	Correct	Question supplied, procedure designed by learners.
B	Inverted operation	That makes it a confirmation lab.
C	Confusion set	That is open inquiry, not guided.
D	Adjacent category	That is structured inquiry, one level lower.

9. Answer D Concept 6 • transfer

In a lab, the teacher provides the question and the procedure, and learners do not know the expected result. What level of inquiry is this?

Target word: name the level **Decides:** Question and procedure given; result unknown.

A	Adjacent category	Confirmation also gives the expected result.
B	Confusion set	Guided inquiry has learners design the procedure.
C	Inverted operation	Open inquiry has learners supply the question too.
D	Correct	Learners find the result using a given question and procedure.

10. Answer A Concept 7

To check understanding, a teacher candidate asks learners to draw what salt (NaCl) looks like dissolved in water. ~~The class had just finished a lab using iodized table salt from a grocery store.~~ Which drawing shows correct understanding?

Target word: correct particle model of dissolved NaCl **Decides:** Ionic compounds dissociate into separate, hydrated ions.

A	Correct	Shows dissociation and hydration.
B	Adjacent category	The particles are charged ions, not neutral atoms.
C	Confusion set	Treats the ionic solid as if it dissolved as molecules.
D	Dropped element	Leaves out the mixing that makes a solution uniform.

11. Answer C Concept 7 • transfer

Which drawing correctly shows table sugar (sucrose) dissolved in water?

Target word: particle model of a molecular solute **Decides:** Molecular solutes stay as whole molecules.

A	Absolute language	The sugar is still present; it is just spread out.
B	Adjacent category	Dissolving does not break covalent bonds.
C	Correct	Sucrose separates from the crystal but its molecules stay intact.
D	Confusion set	Sucrose is covalent and does not dissociate into ions.

12. Answer B Concept 8

The lesson objective reads: "Learners will predict the products of double-replacement reactions using solubility rules." ~~The candidate's mentor prefers short assessments that fit on one page.~~ Which assessment task aligns best?

Target word: task matching the objective's verb **Decides:** The verb is predict, using solubility rules.

A	Adjacent category	Recall of a definition is lower than predicting.
B	Correct	Requires the exact performance the objective names.
C	Absolute language	A written task can require prediction and justification.
D	Confusion set	Related chemistry skill, but not the one in the objective.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	10	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Elicit prior ideas before teaching	Surface learners' existing ideas with a probe before instruction so teaching can target them.	Lecturing first and fixing misconceptions only if they happen to come up.	"Before starting the unit..."
2	Formative vs. summative use	Formative assessment is used during learning to adjust teaching; summative assessment judges learning at the end.	Labeling by format (quiz, ticket) instead of by how the results are used.	What the teacher does with the results.
3	Evaluating models and simulations	Choose resources that are scientifically accurate at the particle level, and name the limits of any model you use.	Picking a resource because it is engaging or free, or presenting a model as literal truth.	Two resources compared, one with a subtle science error.
4	5E sequence: explore before explain	After Engage, learners Explore the phenomenon before the teacher Explains the formal concept.	Giving the explanation right after the hook.	"What should come next?" after an opening demo.
5	Worked examples and fading	Novices learn multistep procedures best from worked examples, then partially completed problems, then independent practice.	Assigning many independent problems first so learners "figure it out."	Novices struggling with a multistep calculation such as stoichiometry.
6	Levels of inquiry in lab resources	Confirmation: question, procedure, and answer given; structured: question and procedure given; guided: question given, learners design the procedure; open: learners pose the question.	Calling any hands-on lab inquiry, or skipping straight to open inquiry.	Who supplies the question, the procedure, and the expected result.
7	Particle diagrams as assessment	Ionic solids dissolve into separated, hydrated ions; molecular solids dissolve as intact molecules. Learner drawings reveal which model they hold.	Accepting drawings of intact NaCl pairs, or sugar split into ions.	"Which drawing shows...dissolved in water?"
8	Aligning assessment to the objective	The task must ask learners to do what the objective's verb says, at the same level of thinking.	Testing recall of definitions when the objective says predict or explain.	An objective with an action verb and a list of candidate tasks.

Course scope (public description): Secondary Chemistry Curriculum synthesizes the core content of the Secondary Chemistry Program, preparing candidates to apply education theory and teaching approaches to the chemistry content areas. This course aligns with the areas covered by the Chemistry Praxis exam, ensuring candidates demonstrate both a deep understanding of the subject matter and the ability to effectively deliver content in a secondary classroom setting. Through evidence-based methods of learning science, candidates will evaluate learning experiences and resources that effectively convey chemistry concepts to secondary students. Topics include atomic structure, chemical bonding, energy relationships, chemical reactions, solutions, and the role of science and engineering in society. There are no prerequisites for this course.